



Exit Exam Tutorial

Part 2: Fundamental Database Management Systems

Episode 4: Functional Dependency and Normalization

2.4 Functional Dependency and Normalization

- Database normalization is a series of steps followed to obtain a database design that allows for consistent storage and efficient access of data in a relational database.
- These steps reduce data redundancy and the risk of data becoming inconsistent.
- **NORMALIZATION** is the process of identifying the logical associations between data items and designing a database that will represent such associations but without suffering the update anomalies which are;
 1. Insertion Anomalies
 2. Deletion Anomalies
 3. Modification Anomalies

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- Normalization may reduce system performance since data will be cross referenced from many tables.
- Thus de-normalization is sometimes used to improve performance, at the cost of reduced consistency guarantees.
- Normalization normally is considered as good if it is lossless decomposition.
- Mnemonic for remembering the rationale for normalization could be the following:
 1. No Repeating or Redundancy: no repeating fields in the table.
 2. The Fields Depend Upon the Key: the table should solely depend on the key.
 3. The Whole Key: no partial key dependency.
 4. And Nothing But The Key: no inter-data dependency.

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Pitfalls of Normalization

- Requires data to see the problems
- May reduce performance of the system
- Is time consuming,
- Difficult to design and apply and
- Prone to human error

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The type of problems that could occur in insufficiently normalized table is called update anomalies which includes;

- 1. INSERTION ANOMALIES:** failure to place information about a new database entry into all the places in the database where information about that new entry needs to be stored.
- 2. DELETION ANOMALIES:** failure to remove information about an existing database entry when it is time to remove that entry.
- 3. MODIFICATION ANOMALIES:** involves changing some value of the attribute of a table.

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- **Data Dependency:** The logical association between data items that point the database designer in the direction of a good database design are referred to as determinant or dependent relationships.
- In general, a functional dependency is a relationship among attributes.
- In relational databases, we can have a determinant that governs one other attribute or several other attributes.
- **Partial Dependency:** If an attribute which is not a member of the primary key is dependent on some part of the primary key (if we have composite primary key) then that attribute is partially functionally dependent on the primary key.

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- **Full Dependency:** If an attribute which is not a member of the primary key is not dependent on some part of the primary key but the whole key (if we have composite primary key) then that attribute is fully functionally dependent on the primary key.
- **Transitive Dependency:** In mathematics and logic, a transitive relationship is a relationship of the following form: "If A implies B, and if also B implies C, then A implies C."

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2.4.1 Normalization

1. FIRST NORMAL FORM (1NF): Requires that all column values in a table are atomic (e.g., a number is an atomic value, while a list or a set is not).

- Definition of a table (relation) in 1NF

- There are no duplicated rows in the table/unique identifier.
- Each cell is single-valued (i.e., there are no repeating groups).
- Entries in a column (attribute, field) are of the same kind.

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2. SECOND NORMAL FORM (2NF): No partial dependency of a non key attribute on part of the primary key.

- This will result in a set of relations with a level of Second Normal Form.
- Any table that is in 1NF and has a single-attribute (i.e., a non-composite) key is automatically also in 2NF.
- Definition of a table (relation) in 2NF
 - It is in 1NF
 - If all non-key attributes are dependent on all of the key. i.e. no partial dependency.
- Since a partial dependency occurs when a non-key attribute is dependent on only a part of the (composite) key, the definition of 2NF is sometimes phrased as, "A table is in 2NF if it is in 1NF and if it has no partial dependencies."

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3. THIRD NORMAL FORM (3NF): Eliminate Columns Not Dependent On Key - If attributes do not contribute to a description of the key, remove them to a separate table.

- This level avoids update and delete anomalies.
- Definition of a table (relation) in 3NF
 - It is in 2NF
 - There are no transitive dependencies between attributes.

4. BOYCE-CODD NORMAL FORM (BCNF): Isolate Independent Multiple Relationships - No table may contain two or more 1:n or N:M relationships that are not directly related.

- A table is in BCNF if it is in 3NF and if every determinant is a candidate key.

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5. FOURTH NORMAL FORM (4NF): Isolate Semantically Related Multiple Relationships.

- There may be practical constraints on information that justify separating logically related many-to-many relationships.
- A table is in 4NF if it is in BCNF and if it has no multi-valued dependencies.

6. FIFTH NORMAL FORM (5NF): A model limited to only simple (elemental) facts, as expressed in ORM.

- A table is in 5NF, also called "Projection-Join Normal Form" (PJNF), if it is in 4NF and if every join dependency in the table is a consequence of the candidate keys of the table.

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7. DOMAIN-KEY NORMAL FORM (DKNF): A model free from all modification anomalies.

- A table is in DKNF if every constraint on the table is a logical consequence of the definition of keys and domains.

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2.4.2 Physical Database Design Methodology for Relational Database

- The Logical database design is concerned with the what.
- The Physical database design is concerned with the how.
- Physical database design is the process of producing a description of the implementation of the database on secondary storage.
- It describes the base relations, file organization, and indexes used to achieve effective access to the data along with any associated integrity constraints and security measures.
- Physical design describes the base relation, file organization, and indexes used to achieve efficient access to the data, and any associated integrity constraints and security measures.
- Sources of information for the physical design process include global logical data model and documentation that describes model.

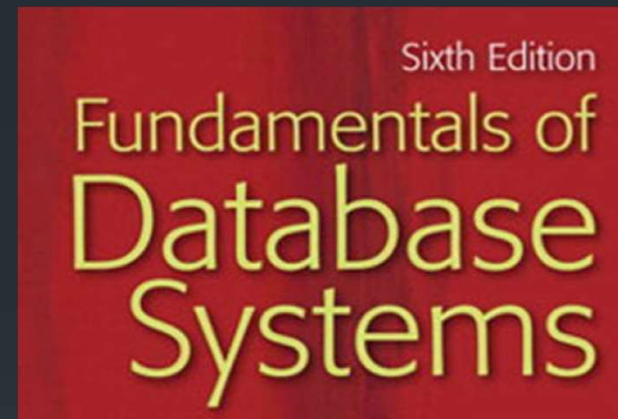
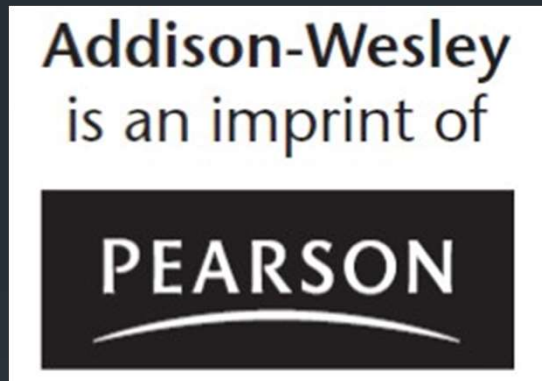
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- Describes the storage structures and access methods used to achieve efficient access to the data
- Knowledge of the DBMS that is selected to host the database systems, with all its functionalities, is required since functionalities of current DBMS vary widely.

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PEARSON

Database Systems

*A Practical Approach to Design, Implementation,
and Management*

SIXTH EDITION

TOPICS AND THE CONCEPTS:

Functional Dependency
Partial Dependency
Full Dependency
Transitive Dependency
First Normal Form
Second Normal Form
Third Normal Form
Boyce-Codd Normal Form
Fourth Normal Form
Fifth Normal Form
Domain-Key Normal Form
Logical and Physical Database for Relational Database

REFERENCES:

Fundamental Database Management Systems (6th Edition) by Ramez Elmasri, Shamkant B. Navaathe
Database Systems: A Practical Approach to Design, Implementation, and Management (6th Edition) by
Thomas Connolly, Carolyn Begg

PRESENTED BY:

Mohammed Nebil

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